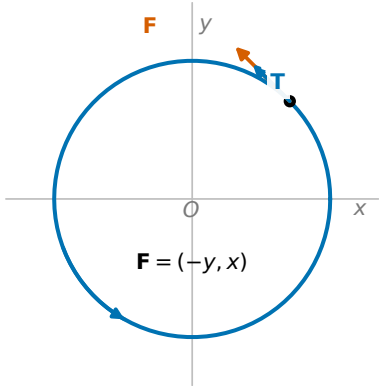
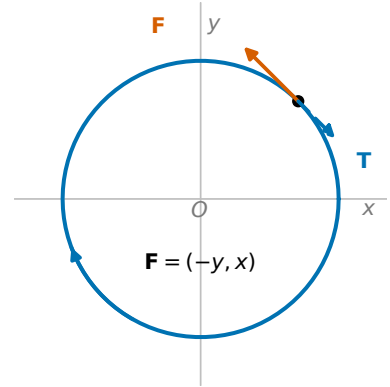
A



$\int_C \mathrm{d}s = 2\pi \qquad \oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = 2\pi$

B



$\int_C \mathrm{d}s = 2\pi \qquad \oint_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = -2\pi$